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An unusual distribution of the *kdr* gene among populations of *Anopheles gambiae* on the island of Bioko, Equatorial Guinea

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Abstract

In West Africa, *Anopheles gambiae* exists in discrete subpopulations known as the M and S molecular forms. Although these forms occur in sympatry, pyrethroid knock-down resistance (*kdr*) is strongly associated with the S molecular form. On the island of Bioko, Equatorial Guinea we found high frequencies of the *kdr* mutation in M form individuals (55.8%) and a complete absence of *kdr* in the S form. We also report the absence of the *kdr* allele in M and S specimens from the harbour town of Tiko in Cameroon, representing the nearest continental population to Bioko. The *kdr* allele had previously been reported as absent in populations of *An. gambiae* on Bioko. Contrary to earlier reports, sequencing of intron-1 of this sodium channel gene revealed no fixed differences between M form resistant and susceptible individuals. The mutation may have recently arisen independently in the M form on Bioko due to recent and intensive pyrethroid application.

Keywords: *Anopheles gambiae*, *kdr*, Bioko, molecular forms.

Introduction

Pyrethroid insecticides are the preferred choice for impregnating bed nets because of their high efficacy, rapid rate of knockdown, residual and strong mosquito excito-repellent properties and low mammalian toxicity (Diabate *et al*., 2002). The use of insecticide-treated bed nets (ITN) offers both individual and community protection against malaria, sometimes reducing morbidity by as much as 50% and global mortality by 20–30% (Binka *et al*., 1996). The spread of pyrethroid resistance in *Anopheles gambiae* populations across Africa is becoming a hindrance to malaria control programs based on ITNs using pyrethroids (Elissa *et al*., 1993). Knockdown resistance is primarily conferred by the knock down resistance (*kdr*) gene (Chandre *et al*., 1999), a single point mutation in domain II of the voltage-gated sodium channel gene – located on the left arm of chromosome 2 in *An. gambiae* (Williamson *et al*., 1996; Ranson *et al*., 2000a). This mutation involves the replacement of leucine by phenylalanine. The addition of the large aromatic side chain of phenylalanine may impose steric hindrance interfering with insecticide binding to the sodium channel thus rendering the individual mosquito resistant (Ranson *et al*., 2000b).

Anopheles gambiae s.s is classified into two molecular forms, Mopti (M) and Savannah (S). The forms exhibit strong assortative mating and are considered by many to be incipient species (Lanzaro & Tripet, 2003). Pyrethroid *kdr* occurs in many populations of *An. gambiae* throughout the species range: in West Africa, *kdr* is strongly associated with the S molecular form of *An. gambiae*. The *kdr* mutation was first identified in the S molecular form of *An. gambiae* and until recently had yet to be detected in the M form, even at sites where the two are sympatric.

Resistance in the S form has been reported from the Ivory Coast (Weill *et al*., 2000; della Torre *et al*., 2001), Ghana (della Torre *et al*., 2001), Nigeria (Awolola *et al*., 2003), Mali (Fanello *et al*., 2003) and Burkina Faso (Diabate *et al*., 2003). Resistance in the M form has only been found in Benin (Weill *et al*., 2000; della Torre *et al*., 2001), Burkina Faso (Diabate *et al*., 2004) and at very low frequency in Ghana (Yawson *et al*., 2004). This suggested that the *kdr* mutation reached the M form only recently through introgression from S form populations (Weill *et al*., 2000). Resistance has not been detected in either form in the Central African Republic

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(Weill *et al*., 2000), Senegal, The Gambia, Guinea, Angola (Gentile *et al*., 2004) or Cameroon (Weill *et al*., 2000; Etang *et al*., 2003; Gentile *et al*., 2004).

The volcanic islands of Equatorial Guinea are tropical, with a hot and humid climate year-round. Bioko Island lies 40 km off the coast of Cameroon in the Bight of Biafra. The island rises steeply to an altitude of 2850 m, with two main peaks, one in the north and one in the south. The southern area is rugged and inaccessible. Recent research on Bioko has included seroparasitological studies of malaria, the *in vitro* responses of *Plasmodium falciparum* to antimalarials, mosquito taxonomic surveys and mosquito species distributions. *An. gambiae* and *Anopheles funestus* are considered the main malaria vectors in Bioko. Malaria is hyperendemic on Bioko and is one of three main causes of morbidity and mortality on the island (Berzosa *et al*., 2002).

Malaria exhibits a stable year round pattern of transmission with seasonal fluctuations. In a study of the *in vitro* response of *P. falciparum* to antimalarial drugs, resistant isolates were found to exhibit interregional differences (Benito *et al*., 1995). Prevalence of malaria infection is 29.8% in Bioko, parasitic prevalence (malaria index) was 26.6%, and the splenic index was 57.0% (Roche *et al*., 1991). These results indicate that Bioko is an area of stable hyperendemic malaria.

Populations of *An. gambiae* on Bioko were found to lack the *kdr* allele between 1998 and 2001, despite the use of pyrethroid-impregnated bed nets (Berzosa *et al*., 2002). The fact that the local population was found to be susceptible justified the implementation of a larger scale ITN program through the Malaria Control Program of Equatorial Guinea. In addition, in 2003 Marathon Oil Corporation and Medical Care Development International joint launched a five year Bioko Island Malaria Control Project that includes indoor residual spraying with deltamethrin, a synthetic pyrethroid, in 25 000 homes in Malabo, Bioko Island (MCDI Newsletter, March, 2004).

We surveyed mosquito populations from six villages near Malabo for the presence of the *kdr* gene using a PCR-based diagnostic (Martinez-Torres *et al*., 1998). We found the *kdr* resistance allele in the M molecular form in Malabo. Previous studies have shown polymorphism in intron 1 and fixed differences between the M and S populations from different locales on the mainland (Weill *et al*., 2000). Weill *et al*. (2000) and Diabate *et al*. (2004) reported that the patterns of polymorphism in the intron suggest introgression of the *kdr* alleles from the S to the M form. In order to determine whether the *kdr* allele arose through introgression from mainland mosquitoes or independently through continuous pyrethroid exposure we examined these polymorphic sites in M and S molecular form samples from Bioko and nearby mainland Cameroon.

Results and discussion

Species and molecular form composition

Mosquitoes were collected from 6 villages within 8 km of Malabo, the capital of Bioko and from the village of Tiko in Cameroon. M and S molecular form diagnostics were performed on 75 *An. gambiae* from Malabo and 68 from Tiko. In Malabo 27.7% of mosquitoes collected were *An. funestus* (n = 33) and 72.3% were *An. gambiae* (n = 86); of these 27 (36%) were S form and 48 (64%) were M form. In Tiko 100% (n = 68) of mosquitoes collected were *An. gambiae*, 14 (21%) were S form and 54 (79%) were M form. Although M and S form individuals were collected from the same sites no M/S hybrids were observed in either Malabo or Tiko. This suggests that if hybrids are generated, it is at a very low frequency. This observation is consistent with what has been reported throughout West Africa (Lanzaro & Tripet, 2003).

Genetic differentiation and gene flow

Pair-wise F_{ST} values, based on eight microsatellite loci, between one island and three mainland populations, as

Table 1. Matrices of pair-wise estimates of genetic divergence (F_{ST}) and significance levels of exact tests of genetic differentiation between and among molecular forms on Bioko Island and three locales in Cameroon – Malantouen, Mutanguene and Tiko

Location and molecular form	Malantouen S	Mutanguene M	Mutanguene S	Bioko M	Bioko S	Tiko M	Tiko S
Malantouen S	0						
Mutanguene M	0.0708**	0					
Mutanguene S	0.0039	0.1024**	0				
Bioko M	0.1085**	0.0383**	0.1299**	0			
Bioko S	0.0582**	0.0937**	0.0558**	0.0699**	0		
Tiko M	0.0766**	0.0074	0.0891**	0.0575**	0.0914**	0	
Tiko S	–0.0148	0.0739**	–0.0027	0.0999**	0.0238*	0.0757**	0

*P < 0.01; **P < 0.001.

well as between molecular forms of *An. gambiae* were calculated (Table 1). All between form F_{ST}s were significant, indicating that gene flow between molecular forms is limited at every location sampled, even where the two occur in sympatry. F_{ST} estimates between Tiko, mainland Cameroon and Bioko Island were high and significant, both between and within forms. Tiko is a harbour town, 75 km across the Bight of Biafra on mainland Cameroon. Although trade and transportation are frequent between Bioko Island, Cameroon and Equatorial Guinea, Tiko represents the nearest continental population to the island. In order to determine whether the Bight of Biafra poses a physical barrier to gene flow, we also included in our analysis two additional locations in Cameroon; Mutanguene (n = 29) and Malantouen (n = 27). The distance from each village to Tiko is 85 km and 345 km, respectively. F_{ST} estimates within form among mainland populations were all low, suggesting the Bight of Biafra is a strong physical barrier to gene flow between populations on Bioko Island and mainland Cameroon.

Frequency and distribution of kdr *allele*

Presence of the *kdr* allele was determined by PCR as described by Martinez-Torres *et al.* (1998). Although the *kdr* allele was not detected in any *An. gambiae* samples from Bioko collected between 1998 and 2001 (Berzosa *et al.*, 2002), we found the *kdr* allele at high frequency in our M form samples from Bioko. Near Malabo, the *kdr* allele was present in 55.8% of the M molecular form (5 R/R, 12 R/S and 2 R/?) but was absent in the S molecular form. The *kdr* allele was also absent in all *An. gambiae* sampled from Tiko (n = 68). The Tiko samples belong to the forest chromosomal form and consist of both the M and S molecular form. A recent study by Berzosa *et al.* (2002) demonstrated that all *An. gambiae* s.s. on Bioko Island belong to the forest chromosomal form.

Origin of kdr *in* An. gambiae *on Bioko*

We analysed a 526 bp region including the *kdr* mutation and a portion of the intron upstream of the *kdr* mutation (Fig. 1). We also sequenced the same region in four mosquitoes collected from Tiko. Weill *et al.* (2000) identified two polymorphic sites at nucleotide positions 702 and 896 (following the notation by Weill *et al.*, 2000) in which M form susceptible individuals carry C-A or C-C combinations and M form resistant along with most S form carry a T-C combination. A recent study by Gentile *et al.* (2004) has shown a correlation between the substitution at site 702 and molecular form, whereby M form resistant individuals and all S form individuals exhibit the thymine substitution and M form susceptible individuals exhibit cytosine at this site. The few exceptions to this trend were thought to have arisen from past introgression events between M and S (Gentile *et al.*, 2004). In this study we found the expected T-C combination at sites 702 and 896, respectively, in M form resistant mosquitoes from Bioko. However, the T-C combination, thought to be characteristic of all S form individuals, but only in M form individuals carrying the *kdr* gene, was also found in M form susceptible mosquitoes on Bioko (Table 2). Therefore we found no differences between susceptible and resistant M form individuals outside of the

Table 2. Polymorphisms in *Anopheles gambiae* s.s. molecular forms collected on the island of Bioko and the port city of Tiko in Cameroon. Unlike previous data (Weill *et al.*, 2000; Diabate *et al.*, 2004), no polymorphism was observed at site 896 and the polymorphisms at site 702 were not associated with molecular form or resistance. Pyrethroid knockdown resistance (r) and susceptibility (s) determined by sequencing

Location and molecular form	Sample size	702	896	*kdr*
Malabo, Bioko				
M form	4	T/T	C/C	r/r
	1	**T/T**	**C/C**	**s/r**
	4	C/T	C/C	s/r
	2	C/C	C/C	s/s
	1	**C/T**	**C/C**	**s/s**
S form	1	C/C	C/C	s/s
	4	T/T	C/C	s/s
	2	C/T	C/C	s/s
Tiko, Cameroon				
M form	3	C/C	C/C	s/s
S form	2	T/T	C/C	s/s



Figure 1. Partial sequence of the sodium channel gene showing the Leu-Phe *kdr* mutation that is associated with pyrethroid knockdown resistance (shaded box), a portion of the 920 bp intron 1 that precedes it and the 57 bp intron 2 (bold). Two positions in intron 1 are known to be polymorphic in *Anopheles gambiae* s.s. (702 and 896), though in this study polymorphism was only found at site 702. Forward and reverse primers are also shown (underlined).

single *kdr* point mutation. The absence of the T-C combination in susceptible M form in West Africa has been used to argue for the introgression of the *kdr* allele from the S form into the M form (Weill *et al*., 2000). Our observations, that the T-C combination is present in susceptible M form individuals on Bioko and the absence of the *kdr* gene in the nearest mainland population in Cameroon, suggests that the emergence of *kdr* resistance in the M population of *An. gambiae* on Bioko occurred as an independent evolutionary event.

The T-C combination in the M form in Bioko and Cameroon most likely represents an ancestral shared polymorphism. The molecular forms are thought to have diverged only very recently and differentiation between them is limited to a few genomic regions. The alternative scenario that the M form acquired the T-C combination through introgression from a susceptible S form cannot be ruled out. However, the *kdr* allele is located in one of the few areas of the genome that appears to resist introgression between forms (Chandre *et al*., 1999; Turner *et al*., 2005). In addition, the scenario of an independent origin of the *kdr* mutation on Bioko is supported by the great physical distance between Bioko Island and the nearest source of *kdr* carrying M form individuals from the mainland, in Benin. Furthermore, our estimates of gene flow between Bioko and the nearest mainland population Tiko, suggest that *An. gambiae* populations on Bioko are to a large extent isolated from mainland populations. Finally, an independent origin of the *kdr* mutation is not without precedent. Based on intron polymorphisms, Diabate *et al*. (2004) concluded that the mutation arose independently in *Anopheles arabiensis.* A separate leucine-serine substitution at the *kdr* site, associated with DDT and pyrethroid resistance, was found in *An. gambiae* from Kenya (Ranson *et al*., 2000a).

It is also important to note that the *kdr* gene was not present on the island prior to 2001, before any large scale application of insecticide (Berzosa *et al*., 2002). In 2001 antimalaria efforts in Malabo increased, with frequent and intense application of pyrethroid aerosols from trucks and by hand (A. Spielman, personal communication, May 11, 2005). By 2002, when our samples were collected, the frequency of the *kdr* gene within the M form population around Malabo was 55.8%, a staggering increase. Other studies have shown a strong increase in the frequency of the *kdr* gene immediately following the implementation of an impregnated bed net program (Ranson *et al*., 2000a; Stump *et al*., 2004), although these increases were not as dramatic as we report here. The unusual distribution of the *kdr* mutation in the M form may have arisen independently and recently on the island of Bioko in response to this intensive series of pyrethroid applications. Although the individual and community protection offered by treated bed nets continue to have a powerful impact on malaria transmission (Curtis *et al*., 2003), the potential of wide spread pyrethroid resistance in *An. gambiae* could be disastrous to control efforts.

Experimental procedures

Mosquito collections

Mosquitoes for the *kdr* analyses were collected from two locations: from urban and peri-urban settings in the city of Malabo, Bioko Island, Equatorial Guinea (3°45′N, 8°47′E) and from the rural areas surrounding the harbour town of Tiko, Cameroon (4°04′N, 9°22′E). Collections were also made from the villages of Malantouen (5°44′N, 12°00′E) and Mutanguene (4°50′N, 9°17′E) in Cameroon for estimates of gene flow based on microsatellite allele frequencies. In Bioko, adult female mosquitoes were collected by hand using aspirators and human bait catches in August and September, 2002. The mosquitoes were sorted to genus using morphological identification and genomic DNA of anophelines was extracted using a DNeasy extraction kit (Qiagen, Valencia, CA, USA). In Cameroon, adult resting females were collected indoors using aspirators in September 2003. DNA was extracted using a standard extraction protocol (Post *et al*., 1992).

Species and form diagnostic

The PCR assay described by Scott *et al*. (1993) was used to identify members of the *An. gambiae* species complex. An additional diagnostic described by Koekemoer *et al*. (2002) was used to identify *An. funestus* from Bioko Island. Molecular identification of forms within *An. gambiae* s.s. was based on the method described by Fanello *et al*. (2002).

Estimation of gene flow

We estimated the amount of genetic differentiation between island and mainland populations using eight *3rd* chromosome microsatellite loci of GT repeats that were isolated and mapped by Zheng *et al*. (1996). The loci were PCR amplified using fluorescent primers and an MJ Research PTC-200 thermal cycler (MJ Research, Watertown, MA). PCR products were mixed with a Genescan (Perkin-Elmer, Norwalk, CT) size standard and run on an ABI 3100 capillary sequencer (Perkin Elmer). The gels were analysed using the ABI PRISM Genescan Analysis Software and Genotyper DNA Fragment Analysis Software (Perkin-Elmer). Arlequin version 2.001 (Schneider *et al*., 2001) available at http://lgb.unige.ch/arlequin was used to calculate F_{ST} values between all pairs of populations.

Kdr *diagnostic*

Detection of the Leu-Phe *kdr* mutation was based on the PCR diagnostic test developed by Martinez-Torres *et al*. (1998) with modifications. 2 μl of extracted genomic DNA was combined with primers Agd1, Agd2, Agd3 and Agd4 to a total reaction volume of 25 μl. PCR conditions included an initial 13 min at 95 °C, 1 minute at 95 °C, 30 s at 48 °C, 30 s at 72 °C for 60 cycles and a final extension step at 72 °C for 10 min. Amplified fragments were analysed by electrophoresis using 1.5% agarose gel.

Sequence determination

Based on results of the PCR diagnostic, 24 M and S form, knockdown resistant and susceptible individuals were selected (19 from

Malabo, 5 from Tiko) for sequence determination. A 602 bp genomic region containing the *kdr* mutation site and a portion of intron 1 was PCR amplified using primers Agd8 (5′-CACAACAAGTACAAAATGTCTCGC-3′) and *kdr*-rev (5′-GCAAGGCTAAGAAAAGGTTAAGCA-3′). The region was trimmed to 526 bp for analysis. PCR fragments were purified using QIAquick PCR purification kit (Qiagen) and sequences were analysed using an ABI 3100 Genetic Analyser (Foster City, CA, USA). The sequences were aligned using the Clustal-V method (Megalign program, DNASTAR, Inc., Madison, WI, USA).

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